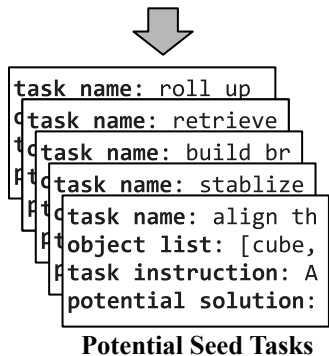
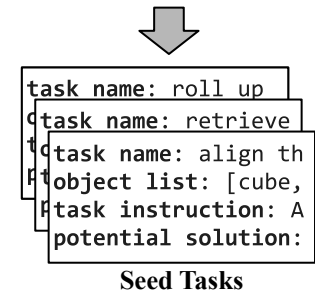


Seed Task Generator



Task Verifier



Metric generation

task name: align three cubes
object list: [cube, cube2,
cube3, long rigid ruler]
task instruction: Align the
three cubes perfectly in a
straight line.
potential solution: Keep
pushing the ruler to the
cubes until all three cubes
touch the ruler.

A Single Seed Task



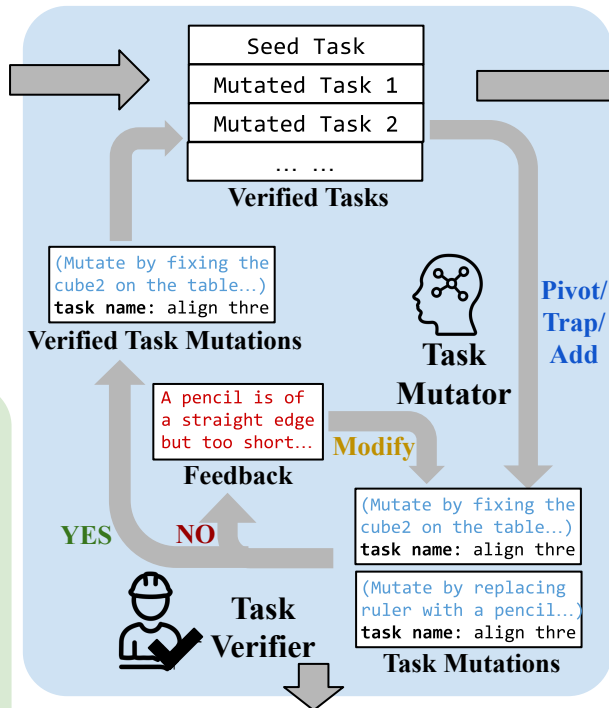
Metric Generator

- ✓ All objects are on the table
- ✓ Three cubes are collinear
- ✓ Robot isn't holding any object

```
def is_completed(objs_info) ->  
bool:  
    cube = objs_info["cube"]  
    ...  
    return on_same_line(cubes)
```

Metrics Code

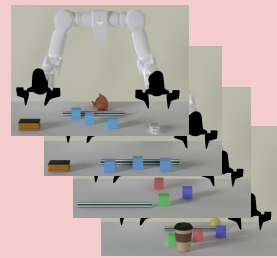
Task Mutation and Verification



Scene Generation



Scene Generator



3D Scene Configs

Mutated Task Family

task name: align th
object list: [cube,
task instruction: A
potential solution:

```
def is_completed(objs_info) ->  
bool:  
    cube = objs_info["cube"]  
    ...  
    return on_same_line(cubes)
```

